

Lecture - 1

Introduction

CoSc3212 – Network and System Administration

Topics

INTRODUCTION

1. What do sysadmins do?
2. Organizations
3. SAGE Classifications and Skills
4. Certifications
5. Principles and First Steps

DESKTOPS

1. Machine Lifecycle
2. Automated Installs
3. Updates
4. Network Configuration

What is a Sysadmin?

○ In a small org, sysadmin can be entire IT staff

- Phone support
- Order and install software and hardware
- Fix anything that breaks from phones to servers
- Develop software

In a large org, sysadmin is part of large IT org

- Specialists instead of “jack of all trades”
- Database admin, Network admin, Fileserver admin, Help desk worker, Programmers, Logistics

What do sysadmins do?

1. Add and remove users.
2. Add and remove hardware.
3. Perform backups.
4. Install new software systems.
5. Troubleshooting.
6. System monitoring.
7. Auditing security.
8. Helping users.

User Management

- ◆ Creating user accounts
 - Consistency requires automation
 - Startup (dot) files
- ◆ Username and UID namespace management
- ◆ Home directory backups and quotas
- ◆ Removing user accounts
 - Consistency requires automation
 - Remove everything, not just homedir and passwd

Hardware Management

- ◆ Adding and removing hardware
 - Configuration, cabling, etc.
 - Device drivers
 - Scheduling downtimes and notifying users
- ◆ Evaluation and purchase
- ◆ Capacity planning
 - How many servers?
 - How much bandwidth, disk space?
- ◆ Data Center management
 - Power, racks, environment (cooling, fire alarm)

Backups

- ◆ Backup strategy and policies
 - Scheduling: when and how often?
 - Capacity planning
 - Location: On-site vs off-site.
- ◆ Installing backup software
- ◆ Performing backups and restores
- ◆ Monitoring backups
 - Checking logs
 - Verifying media

Software Installation

- ◆ Automated consistent OS installs
- ◆ Evaluation of software
- ◆ Finding and building open source software
- ◆ Purchase of commercial software
- ◆ Managing software installations
 - Distributing software to multiple hosts
 - Package management
 - Managing multiple versions of a software pkg
- ◆ Patching and updating software
- ◆ Scheduling downtimes and notifying users

Troubleshooting

◆ Problem identification

- By user notification
- By log files or monitoring programs

◆ Tracking and visibility

- Ensure users know you're working on problem
- Provide an ETA if possible

◆ Finding the root cause of problems

- Provide temporary solution if necessary
- Solve the root problem to permanently eliminate

Performance Monitoring

- ◆ Automatically monitor systems for
 - Problems (disk full, error logs, security)
 - Performance (CPU, mem, disk, network)
- ◆ Log rotation and backups
- ◆ Provides data for capacity planning
 - Convince management of need for hardware

Helping Users

◆ Request tracking system

- Ensures that you don't forget problems.
- Ensures users know you're working on their problem; reduces interruptions, status queries.
- Lets management know what you've done.

◆ User documentation and training

- Acceptable Use Policies
- Document software, hardware (printers), etc.

Qualities of a Successful Sysadmin

◆ Customer oriented

- Ability to deal with interrupts, time pressure
- Communication skills
- Service provider, not system police

◆ Technical knowledge

- Hardware, network, and software knowledge
- Debugging and troubleshooting skills

◆ Time management

- Automate everything possible.
- Ability to prioritize tasks: urgency and importance.

Organizations

USENIX: Advanced Computing Systems Association

LISA: Large Installation System Administration

SAGE: System Administration Guild

SAGE: Goals

1. Advance status of system administration as a profession.
2. Establish standards of professional excellence and recognize those who attain them.
3. Develop guidelines for improving the technical and managerial capabilities of members of the profession.
4. Promote activities that advance the state of the art or the community.

Types of Sites

Small

2-10 computers, 1 OS, 2-20 users.

Midsized

11-100 computers, 1-3 OSes, 21-100 users.

Large

100+ computers, multiples OSes, 100+ users

SAGE Job Descriptions

Novice

UNIX familiarity (CIT 140)

Can explaining simple procedures in writing or verbally, has good phone skills.

Junior

UNIX skills, system administration basics (install, boot, add/remove users) (CIT 370)

Capable of training users in applications and UNIX fundamentals, and writing basic documentation.

SAGE Job Descriptions

Intermediate

Broad system administration knowledge, including setup of common server types.

Understanding of network/distributed computing concepts (directories, authentication, network filesystems).

Ability to automate tasks using sh, perl, etc.

Capable of writing purchase justifications, training users in complex topics, making presentations to an internal audience.

Independent problem solving; self-direction.

SAGE Job Descriptions

Senior

A solid understanding of networking/distributed computing environment concepts; understands principles of routing, client/server programming, the design of consistent network-wide filesystem layouts.

Ability to program in an administrative language (sh, perl), to port C programs from one platform to another, and to write small C programs.

Capable of writing proposals or papers, acting as a vendor liaison, making presentations to customer/client audiences or professional peers.

Ability to solve problems quickly and completely.

Ability to identify tasks which require automation and automate them.

Other Skills

Heterogenous Environments

Integrating multiple-OSes, hardware types, or network protocols.

Site Types

Size variations, distributed sites, local variations.

Hardware

Databases

SQL RDMS

Networking

Complex routing, high speed networks.

Security

Firewalls, authentication, NIDS, cryptography.

Certifications

- ◆ CCNA, CCNP, CCIE
- ◆ cSAGE
- ◆ MCSA and MCSE
- ◆ RHCE
- ◆ Sun Certified System / Network Admin

Principles of SA

Simplicity

- Choose the simplest solution that solves the entire problem.

Clarity

- Choose a straightforward solution that's easy to change, maintain, debug, and explain to other SAs.

Generality

- Choose reusable solutions and open protocols.

Automation

- Use software to replace human effort.

Communication

- Be sure that you're solving the right problems and that people know what you're doing.

Basics First

- Solve basic infrastructure problems before moving to advanced ones.

First Steps

Use a request system.

- Customers know what you're doing.
- You know what you're doing.

Manage quick requests right

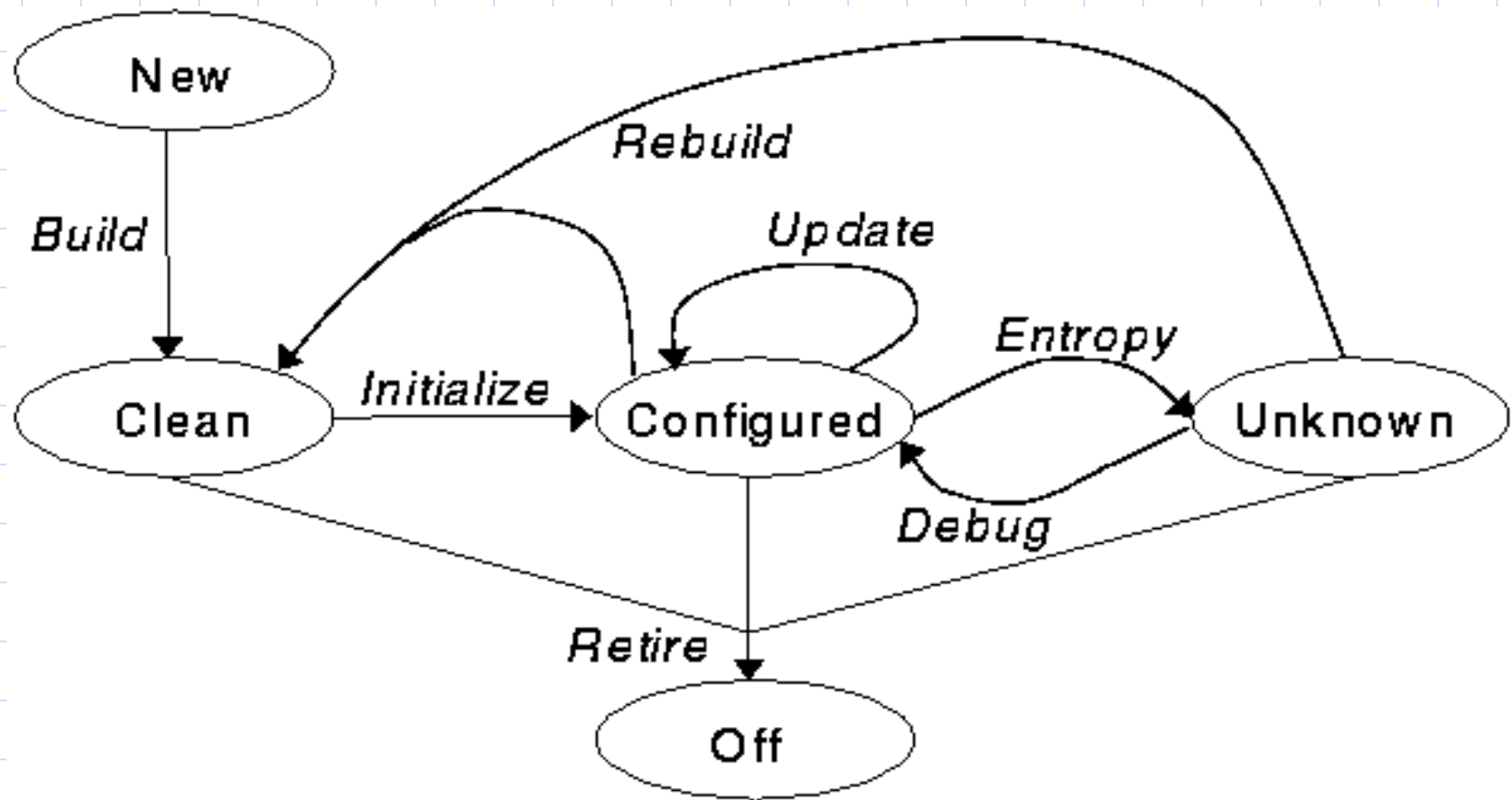
- Handle emergencies quickly.
- Use request system to avoid interruptions.

Policies

- How do people get help?
- What is the scope of responsibility for SA team?
- What is our definition of emergency?

Start every host in a known state.

Desktop Management



States of Machines

New

A new machine

Clean

OS installed, but not yet configured for environment.

Configured

Configured correctly for the operating environment.

Unknown

Misconfigured, broken, newly discovered, etc.

Off

Retired/surplussed

State Transitions

Build

Set up hardware and install OS.

Initialize

Configure for environment; often part of build.

Update

Install new software.

Patch old software.

Change configurations.

Why Automate Installs?

1. Save time.

Boot the computer, then go do something else.

2. Ensure consistency.

No chance of entering wrong input during install.

Avoid user requests due to mistakes in config.

What works on one desktop, works on all.

3. Fast system recovery.

Rebuild system with auto-install vs. slow tapes.

Trusting the Vendor Installation

Always reload the OS on new machines.

- You need to configure the host for your env.
- Eventually you'll reload the OS on a desktop, leaving you with two platforms to support: the vendor OS install and your OS install.
- Vendors change their OS images from time to time, so systems you bought today have a different OS from systems bought 6 months ago.

Install Types

1. Hard Disk Imaging

Duplicate hard disk of installed system.

Advantages: fast, simple.

Disadvantages: need identical hardware, leads to many images, all of which must be updated manually when you make a change

2. Scripted Installs

Installer accepts input from script.

Advantages: flexible, systems can be different

Disadvantages: more effort to setup initially

Auto-Install Features

1. Unattended

Requires little or no human interaction.

2. Concurrent

Multiple installs can be performed at once.

3. Scalable

New clients added easily.

4. Flexible

Configurable to do custom install types.

Auto-Install Components

Boot Component

- Media (floppy or CD)

- Network (PXE)

Network Configuration

- DHCP: IP addresses, netmasks, DNS

Install Configuration

- Media (floppy or CD)

- Network (tftp, ftp, http, NFS)

Install Data and Programs

- Network (tftp, ftp, http, NFS)

PXE

Preboot eXecution Environment

Intel standard for booting over the network.

PXE BIOS loads kernel over network.

Applications

Diskless clients (use NFS for root disk.)

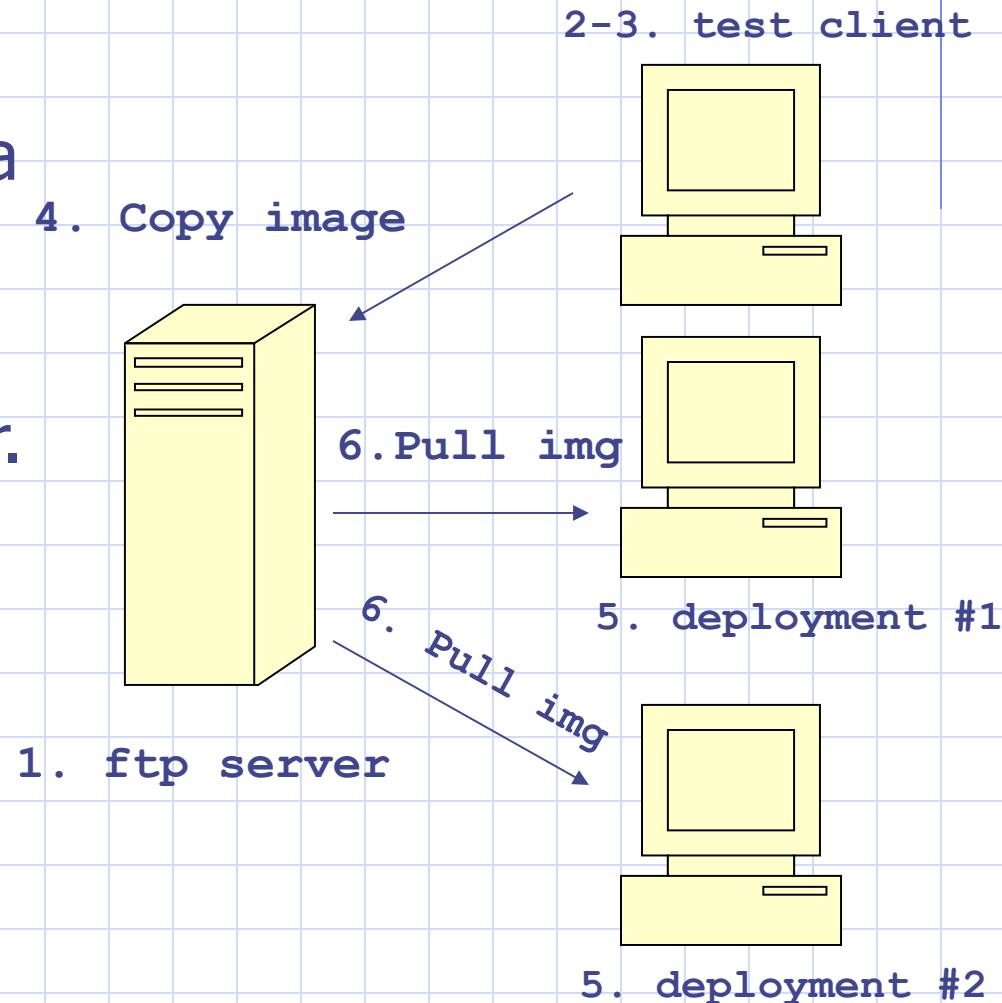
Booting install program.

How it works

1. Asks DHCP server for config (ip, net, tftp.)
2. Downloads pxelinux from tftp server.
3. Boots pxelinux kernel.
4. Kernel uses tftp'd filesystem image or NFS filesystem.

Disk Imaging

1. Setup ftp server.
2. Install OS image on a test client.
3. Verify test client OS.
4. Copy image to server.
5. Boot clients with imaging media.
6. Clients pull image from ftp server.



Using g4u

1. Enable ftp server (service/chkconfig)
2. Download g4u
3. Copy g4u to a floppy disk (or CD)
`cat g4u-2.1-1.fs >/dev/fd0`
4. Boot installed client with floppy disk.
5. Upload image to server.
`uploaddisk your.ftp.server.com filename.gz`
6. Boot blank client with floppy disk.
7. Install image from server.
`slurpdisk your.ftp.server.com filename.gz`

Disk Imaging Tools

- ◆ Acronis TrueImage
- ◆ Clonezilla (free)
- ◆ g4u: Ghost for UNIX (free)
- ◆ Symantec GHOST
- ◆ System Imager (free)

Kickstart Components

Bootable media

- Small bootstrap kernel and filesystem.
- Uses DHCP server to configure system.

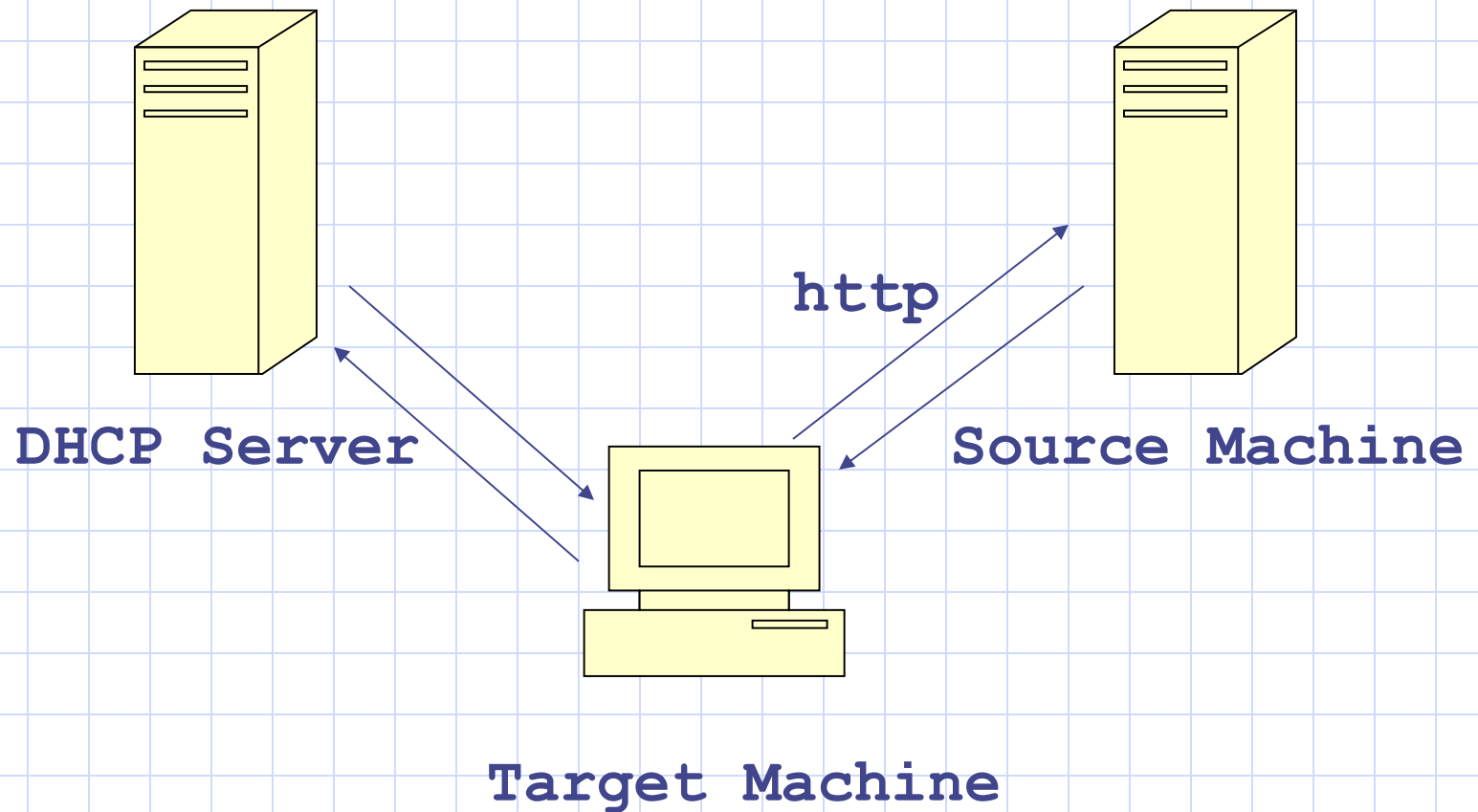
Source machine

- Network server: ftp, http, nfs.
- Kickstart configuration file(s).
- Install files (RPMs).

Target machine

- Machine on which you're installing.
- Boot with bootable media.

Kickstart Components



Source Machine Setup

1. Start network service.
2. Copy install media--for each CD:

```
mount /mnt/cdrom  
cp -var /mnt/cdrom/RedHat  
/usr/local/ks  
umount /mnt/cdrom/
```

3. Create config files.

Store under kickstart subdirectory.

Kickstart Configuration File

Describes desired system configuration.

- Disk partition setup.

- Network configuration.

- Language and other configuration items.

- Package selection.

- Pre- and post-install scripts for customization.

Creating a Kickstart file:

- Original install (located under /root)

- Kickstart Configurator application

- Manually

Kickstart Configurator

The screenshot shows the 'Kickstart Configurator' window with a menu bar containing 'File' and 'Help'. A left sidebar lists configuration categories: 'Basic Configuration' (selected), 'Installation Method', 'Boot Loader Options', 'Partition Information', 'Network Configuration', 'Authentication', 'Firewall Configuration', 'X Configuration', 'Package Selection', 'Pre-Installation Script', and 'Post-Installation Script'. The main area is titled 'Basic Configuration (required)' and contains the following settings:

- Default Language: English (dropdown)
- Keyboard: U.S. English (dropdown)
- Mouse: Generic - 3 Button Mouse (PS/2) (dropdown)
 - ☐ Emulate 3 Buttons
- Time Zone: America/New_York (dropdown)
 - ☐ Use UTC clock
- Root Password: [empty text field]
 - ☒ Encrypt root password
- Language Support: [list box with scroll bar]
 - ☐ Chinese(Mainland)
 - ☐ Chinese(Taiwan)
 - ☐ Czech
 - ☐ Danish
 - ☐ Dutch
 - ☐ English
 - ☐ French
 - ☐ German
 - ☐ Icelandic
- ☒ Reboot system after installation
- ☐ Perform installation in text mode (graphical is default)
- ☐ Perform installation in interactive mode

Configuration Options

auth

crypt, md5, nis, ldap, smb, krb5

network and firewall

DHCP, static, firewall configuration

part

Create disk partitions: size, maxsize, grow.

c.f. autopart, clearpart, log, raid.

rootpw

xconfig

packages

Performing a Kickstart Install

1. Boot with install media

RHEL CD #1

Bootable Kickstart media

2. Specify Kickstart file location

Web: ks=http:// <server>/ <path>

NFS: ks=http:// <server>/ <path>

Floppy: ks=floppy

PXE: ks

Auto-Install Tools

- ◆ DrakX: Mandriva Linux
- ◆ FAI, Preseed: Debian Linux
- ◆ Jumpstart: Solaris
- ◆ Kickstart: Red Hat Linux

Software Update Difficulties

No physical access

- Update process should work w/o physical access.

Host may not be in known state

- Prior updates may or may not have happened.
- Sysadmins or users may have reconfigured.

Hosts may not be there

- Portable computers may not be on your network when you're updating systems.

Host may have live users

- Some updates require no user access or reboots.

One, Some, Many

Failed updates break someone's machine.

- Vendor hasn't tested updates in your env.

One, some, many process mitigates risks

- One: Test update on one system first.
- Some: Test update on group of test systems that are representative of the target systems.
- Many: Schedule update for a time that limits disruption and update user systems.

Network Configuration

What's so bad about manual net settings?

- It's only an IP address and netmask.
- What happens if you need to renumber?

Use DHCP instead of manual settings

- Make all changes on a single server.
- Easy to change network settings for entire net.
- DHCP can assign static IPs as well as dynamic.

Key Points

Being a Sysadmin

- Customer-oriented, technical knowledge, time.
- Basics: request system, known host state, policies.

Desktop Lifecycle

- New, clean, configured, unknown states.

Automated Installs

- Why: consistency, fast recovery, saves time.
- Install types: imaging vs. scripted.
- Components: boot, network, config, data.

One, some, many Approach to Updates.

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